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KnowSelf

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Enabling agents to autonomously regulate knowledge utilization through situational self-awareness

Current Agent Training Uses "Flood Irrigation"

Existing approaches indiscriminately inject knowledge regardless of situational needs, leading to pattern collapse and high inference costs.

Trajectory Imitation

ReAct

Blindsly copies gold trajectories without situational evaluation. Fails on novel inputs.

Trial-and-Error

Reflexion

Relies on external feedback for every correction. Fragile to feedback quality.

Knowledge-Augmented

KnowAgent, WKM

100% knowledge injection at every step. Wasteful and suboptimal.

VS

Humans possess situational self-awareness — the metacognitive ability to know when we need help.

Three Situational Thinking Modes

KnowSelf identifies three cognitive situations based on the agent's ability at each decision step.

⚡ Fast Thinking

Agent directly produces the correct action with minimal deliberation.

`[FAST]`

🔄 Slow Thinking

Agent initially produces incorrect action but self-corrects through reflection.

`[REFLECT]`

📚 Knowledgeable Thinking

Agent cannot produce correct action even after reflection. Must consult external knowledge.

`[KNOW]`

🤔 Can agent produce correct action?

Yes	No
→	→
Correct on first try?	Reflection helps?
Yes No	Yes No

KnowSelf Framework

A three-stage pipeline: Data Construction → Two-Stage Training → Self-Aware Inference

1 Data Construction

Agent self-explores environments. Heuristic criteria classify each step:

- Action matches gold → Fast
- Reflection fixes it → Slow
- Neither works → Knowledge

`[FAST], [REFLECT], [KNOW] tokens inserted`

→

2 Two-Stage Training

Sequential fine-tuning approach:

- **SFT** — Supervised fine-tuning for initial self-awareness
- **RPO** — Reward-based Policy Optimization refines token generation

`SFT → RPO pipeline`

→

3 Self-Aware Inference

At deployment, special tokens trigger:

- Direct action execution
- Reflection cycle
- Knowledge base retrieval

`Autonomous cognitive resource regulation`

KnowSelf Correctly Handles Tasks Where Baselines Fail

Qualitative comparisons on ALFWorld tasks demonstrate KnowSelf's situational reasoning.

Task: Put a hot mug in cabinet

`Heat + Put`

OpenAI-O1 Incorrectly tries to take an egg from the microwave after opening it.

KnowSelf Uses [REFLECT] token to realize it should heat the mug using the open microwave first.

Key: KnowSelf's reflection mechanism enables multi-step planning correction.

Task: Put a saltshaker in drawer

`Put + Place`

DeepSeek-R1 Reflects that drawer is open, tries to close it first (fails on this interpretation).

KnowSelf Applies [KNOW] token to place the object without removing unrelated items.

Key: Knowledge tokens prevent unnecessary intermediate actions.

ALFWorld Results: KnowSelf Achieves 84.33% with Llama-8B

Success rates (%) across six ALFWorld task types. Know% indicates external knowledge usage.

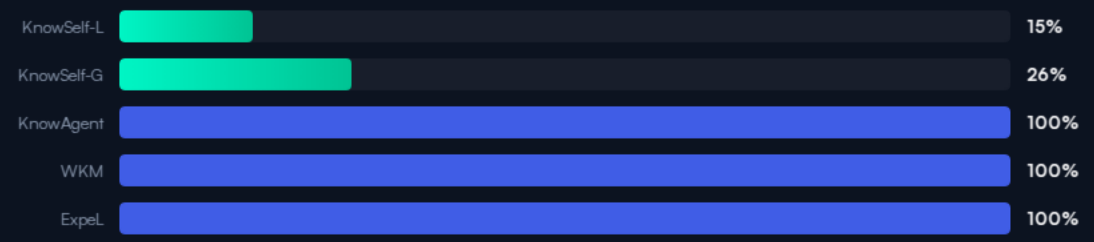
Success rates (%) across ALFWorld task types. Best results in bold.

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	**KnowSelf**	**15%**	**91.67**	**87.10**	**91.30**	**85.71**	**77.78**	**64.71**	**84.33**
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	**KnowSelf**	**26%**	**87.50**	**93.55**	**73.91**	**76.19**	**83.33**	**52.94**	**79.85**

Minimal Knowledge, Maximum Impact

KnowSelf uses only 15—26% knowledge queries while outperforming methods using 100%.

Knowledge Usage Comparison



Overall Accuracy (All Tasks)

Key Efficiency Insight

6x less knowledge usage than full-knowledge methods

KnowSelf-Llama-8B (84.33%) surpasses GPT-4o ExpeL (79.85%) and matches GPT-4o Reflexion (85.07%) using only 15% of the knowledge queries. Even Gemma-2B with KnowSelf (79.85%) matches GPT-4o ExpeL at just 26% knowledge usage.

Key Takeaways

Self-Awareness Is the Missing Piece in Agent Planning

1. **Indiscriminate Knowledge Injection Is Wasteful**
   Methods using 100% knowledge often underperform selective approaches — more knowledge isn't always better.

2. **Self-Awareness Is Learnable**
   Agents can learn situational self-awareness via special tokens with lightweight heuristic labeling — no expensive human annotation needed.

3. **Llama-8B Outperforms GPT-4o Baselines**
   KnowSelf with Llama-8B (84.33%) surpasses GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%).

4. **Minimal Knowledge, Maximum Performance**
   Only 15—26% knowledge usage needed vs. 100% for competing methods — 6× efficiency improvement.

5. **Generalizes Across Model Scales**
   Gemma-2B reaches 79.85% with KnowSelf, matching GPT-4o ExpeL — the paradigm scales down effectively.